

Recap MA - Differentiation

LINEAR ALGEBRA

Def: A vector space over $\mathbb{R}$ is an abelian group V together with a map $\cdot: \mathbb{R} \times V \rightarrow V$, $(\lambda, v) \mapsto \lambda v$ satisfying

$$(i) \lambda(v_1 + v_2) = \lambda v_1 + \lambda v_2$$

$$(ii) (\lambda_1 + \lambda_2)v = \lambda_1 v + \lambda_2 v$$

$$(iii) (\lambda\mu)v = \lambda(\mu v)$$

$$(iv) 1 \cdot v = v$$

$$\forall \lambda, \mu, \lambda_1, \lambda_2 \in \mathbb{R}, \forall v_1, v_2 \in V$$

A subgroup $U \subseteq V$ is a linear subspace if $\lambda u \in U \quad \forall \lambda \in \mathbb{R}, \forall u \in U$.

Def: let V, W be $\mathbb{R}$ -vector spaces. An $\mathbb{R}$ -linear map is a group homomorphism $f: V \rightarrow W$ st $f(\lambda v) = \lambda f(v)$.

Recall: $\text{Ker } f = \{v \in V : f(v) = 0\} \subseteq V$; $\text{Im } f = \{f(v) : v \in V\} \subseteq W$ are linear subspaces.

Theorem: $V/\text{Ker } f \cong \text{Im } f$ for any linear map $f: V \rightarrow W$.
 $\nwarrow$ quotient or factor group.

• If $(b_1, \dots, b_m)$ is an ordered sequence of vectors in V , it defines a linear map

$$b: \mathbb{R}^m \rightarrow V, \quad b(x_1, \dots, x_m) = x_1 b_1 + \dots + x_m b_m$$

Def: The vectors $(b_1, \dots, b_m)$...

1) are linearly independent when b is injective

2) span V when b is surjective

3) form a basis when b is a linear isomorphism, i.e.,

every vector $v \in V$ can be written in a unique way as

$$v = x_1 b_1 + \dots + x_m b_m,$$

and $(x_1, \dots, x_m)$ are the coordinates of v in such a basis.

In this case, $\dim V = m$.

Proposition: let $f: V \rightarrow W$ be a linear map. Then

$$\boxed{\dim V = \dim \text{Ker } f + \dim \text{Im } f}$$

Corollary: let $f: V \rightarrow W$ be a linear map, with $\dim V = \dim W < \infty$. Then
isomorphism $\Leftrightarrow f$ injective $\Leftrightarrow f$ surjective.

- (Matrix of a linear map) : Let $f: V \rightarrow V'$ be a linear map and let $(b_1, \dots, b_m)$ basis for V and $(b'_1, \dots, b'_n)$ basis for V' . Then f is completely determined by the matrix of f (wrt those bases):

$$A = (a_{ij}) = \begin{pmatrix} f(b_1) & \dots & f(b_m) \\ \vdots & & \vdots \end{pmatrix}$$

where $f(b_j) = a_{1j}b'_1 + \dots + a_{nj}b'_n$.

Proposition: If A is the matrix of $f: V \rightarrow V'$ as before, then

$$\boxed{\dim \operatorname{Im} f = \operatorname{rank} A}$$

and therefore

$$\boxed{\dim \operatorname{Ker} f = \#(\text{columns of } A) - \operatorname{rank} A}$$

- (Linear equations): Solving a system of linear equations

$$\left. \begin{array}{l} a_{11}x_1 + \dots + a_{1m}x_m = c_1 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_m = c_m \end{array} \right\}$$

is equivalent to obtaining the points of the level set $f^{-1}(c)$, where $f: \mathbb{R}^m \rightarrow \mathbb{R}^m$ is determined by $A = (a_{ij})$ and $c = \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} \in \mathbb{R}^m$.

To solve such a system, it is enough to find a single solution $x_0 \in f^{-1}(c)$, since the rest of solutions will be given by $x_0 + \operatorname{Ker} f$.

- Sometimes the level sets $f^{-1}(c)$ can be thought as the graph of a linear map.

Lemma (Linear version of Implicit FT): Let $L: V \times W \rightarrow W$ be linear and $c \in W$. Then $L^{-1}(c)$ is the graph of a linear map $F: V \rightarrow W$ (ie $L^{-1}(c) = \operatorname{graph} F \subseteq V \times W$) $\iff L \circ i_2$ is invertible.

- Note that the matrix of a linear map $L: V \times W \rightarrow W$ will be given by a block matrix $(A \mid B)$, where A is the part corresponding to V and B to W . In particular, B is a square matrix. The requirement that $L \circ i_2$ is invertible amounts to B being invertible.

DIFFERENTIATION

Def: A normed vector space is a real vector space V together with a function

$$\|\cdot\|: V \rightarrow [0, \infty)$$

called norm satisfying the following properties:

$$1) \|v\| \geq 0 \quad \forall v \in V, \text{ and } \|v\| = 0 \iff v = 0$$

$$2) \|\lambda v\| = |\lambda| \|v\| \quad \forall \lambda \in \mathbb{R}, v \in V$$

$$3) \|v+w\| \leq \|v\| + \|w\| \quad \forall v, w \in V$$

$$\underline{\text{Ex:}} \text{ On } \mathbb{R}^n, \|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2}; \quad \|x\|_1 = |x_1| + \dots + |x_n|, \\ \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}.$$

An open ball on $(V, \|\cdot\|)$ is a subset $B_r(c) = \{x \in V: \|x-c\| < r\}$.
 $P \subset V$ is open if it is an arbitrary union of open balls. S is closed if $S^c = V \setminus S$ is open. $\overline{B}_r(c) = \{x \in V: \|x-c\| \leq r\}$ is a closed ball.

Def: $f: P \subset V \rightarrow W$ a map between normed spaces is continuous at $p \in P$ if $\forall \varepsilon > 0 \exists \delta > 0: \|x-p\| < \delta \Rightarrow \|f(x) - f(p)\| < \varepsilon$.

Proposition: $f: P \subset V \rightarrow W$ is continuous everywhere $\iff f^{-1}(U)$ open $\forall U \subset W$ open

Def: Let $L: V \rightarrow W$ a linear map between normed spaces.

The operator norm of L is $\|L\| = \max_{\|v\|=1} \|L(v)\|$. This is a norm on the vector space $\text{Hom}(V, W) = \{\text{linear maps } V \rightarrow W\}$.

• Key property: $\|L(v)\| \leq \|L\| \|v\|$

• (Limits): Let $g: P \subset V \rightarrow W$ be a map between normed spaces, let $p \in P$ and $q \in W$. We say that g is the limit of g at p , and we write $\lim_{x \rightarrow p} g(x) = q$, if

$$\forall \varepsilon > 0 \exists \delta > 0: \|x-p\| < \delta \Rightarrow \|g(x) - q\| < \varepsilon$$

Remarks: 1) A limit might not exist (but if it does, it is unique).

2) It is straightforward that g continuous at $p \iff \lim_{x \rightarrow p} g(x) = g(p)$.

* Definition: Let $P \subset V$ be an open subset and $f: P \subset V \rightarrow W$ between normed spaces. We say that f is differentiable at $p \in P$ if there exists a (necessarily unique) linear map $f'(p): V \rightarrow W$ st

$$\lim_{v \rightarrow 0} \frac{f(p+v) - f(p) - f'(p)v}{\|v\|} = 0.$$

Remarks: 1) The definition does not depend on the choice of norm (as all norms on a fin. dim vector space are equivalent).

2) For $V=W=\mathbb{R}$, the previous limit recovers the usual definition of the derivative from calculus: $f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$.

Definition: Let $f: P \subset V \rightarrow W$ bet. normed spaces. The directional derivative of f along the vector $v \in V$ at $p \in P$ is

$$\partial_v f(p) := \lim_{h \rightarrow 0} \frac{f(p+hv) - f(p)}{h}$$

If $(b_1, \dots, b_m)$ is a basis for V , and $(x_1, \dots, x_m)$ are the coordinate system determined by that basis, then the i -th partial derivative of f is $\partial_{b_i} f(p)$. If $W=\mathbb{R}$, then this is denoted as $\frac{\partial f}{\partial x_i}(p)$.

For general W , if $f = (f_1, \dots, f_m): P \subset V \rightarrow W$ (after a choice of basis for W), then $\partial_{b_i} f(p) = \left(\frac{\partial f_1}{\partial x_i}(p), \dots, \frac{\partial f_m}{\partial x_i}(p) \right)$.

• With more generality,

Lemma: $\partial_v f(p) = f'(p)v$, for any $v \in V$.

*Proposition: Let $f: P \subset V \rightarrow W$ be differentiable at p and fix bases on V and W so that $(x_1, \dots, x_m)$ are coordinates in P and $f = (f_1, \dots, f_m)$.

Then the matrix of $f'(p)$ wrt these basis is given by

$$f'(p) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(p) & \dots & \frac{\partial f_1}{\partial x_m}(p) \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(p) & \dots & \frac{\partial f_m}{\partial x_m}(p) \end{pmatrix}$$

and it is usually called the Jacobian matrix.

Definition: A function $f: P \subset V \rightarrow W$ is C^1 if all $\frac{\partial f_i}{\partial x_j}$ exist and they are continuous on P .

*Proposition (Chain rule): Let $P \subset V \xrightarrow{f} Q \subset W \xrightarrow{g} X$ be maps, let f be diff. at $p \in P$ and let g be diff at $f(p)$. Then $g \circ f$ is diff at $p \in P$ and

$$(g \circ f)'(p) = g'(f(p)) \circ f'(p)$$

Proposition (Properties of the derivative)

- 1) $f \in C^1 \Rightarrow f$ diff $\Rightarrow f$ cont.
- 2) f const $\Rightarrow f$ diff & $f'(p) = 0 \quad \forall p \in P$
- 3) f linear $\Rightarrow f$ diff & $f'(p) = f \quad \forall p \in P$

4) $f = (f_1, \dots, f_m)$ diff $\Leftrightarrow f_1, \dots, f_m$ diff, and in that case
 $f'(p) = (f'_1(p), \dots, f'_m(p))$.

Proposition: f is C^1 on $P \Leftrightarrow f$ is continuously differentiable on P , i.e.,
 $f': P \rightarrow \text{Hom}(V, W)$, $p \mapsto f'(p)$ is continuous.

• let $f: P \subset V \rightarrow \mathbb{R}$ be a scalar function. If it exists, denote

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(p) := \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right)(p).$$

When they all exist and are continuous, then f is said to be C^2 on P .

Theorem (Schwarz): Let $f: P \subset V \rightarrow \mathbb{R}$ be C^2 . Then taking partial derivatives commutes:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

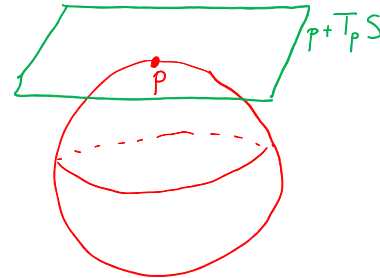
• (Notions of tangent space): We can consider the tangent space of two different objects:

1) Tangent space to a level set: If $S = f^{-1}(c)$ for some $f: P \subset V \rightarrow W$,
 then the tangent space to S at $p \in S$ is defined as

$$T_p S := \text{Ker } f'(p)$$

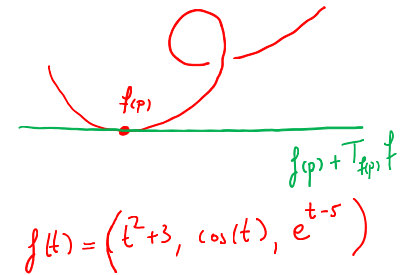
2) Tangent space to the image of a function: If $f: P \subset V \rightarrow W$
 (usually injective) then the tangent space to $f(P)$ at $f(p) \in W$ is

$$T_{f(p)} f := \text{Im } f'(p)$$



$$S = f^{-1}(1),$$

$$f(x, y, z) = x^2 + y^2 + z^2$$



$$f(t) = (t^2 + 3, \cos(t), e^{t-5})$$

MAIN THEOREMS OF MATHEMATICAL ANALYSIS

Def: A contraction is a map $\phi: X \subset V \rightarrow X \subset V$, V normed space, st

$$\|\phi(x) - \phi(y)\| \leq \alpha \|x - y\|$$

for some $0 \leq \alpha < 1$, $\forall x, y \in X$.

* Theorem (Banach fixed point): Let $D = \overline{B}_r(c)$ be a closed ball in a normed space V . Then every contraction $\phi: D \rightarrow D$ has a unique fixed point, i.e. a unique $z \in D$ st $\phi(z) = z$.

• Let $f: P \subset V \rightarrow V$ be a map. A natural question to ask is whether f has an inverse, i.e. if $\exists f^{-1}$ st $f \circ f^{-1} = \text{Id}$, $f^{-1} \circ f = \text{Id}$. In general this is a pretty hard question to answer. An easier question to ask is whether f is locally invertible, i.e. if around some point $z \in P$ there is a neighbourhood $A \subset P$ st $f|_A: A \rightarrow f(A)$ is a bijection. The following theorem gives a sufficient condition for the existence of such a neighbourhood.

* Theorem (Inverse function): Let $f: P \subset V \rightarrow V$ be a C^1 function defined on an open subset P , and let $z \in P$.

If $f'(z)$ is invertible, i.e. if $\det \left(\frac{\partial f_i}{\partial x_j}(z) \right) \neq 0$, then there exists an open neighbourhood $A \subset P$ of z such that $f|_A: A \rightarrow f(A)$ has a C^1 inverse f^{-1} . Furthermore, its derivative is

$$\boxed{(f^{-1})'(f(a)) = f'(a)^{-1}}$$

for $a \in A$.

• The theorem ensures an inverse (typically for a coordinate transformation, e.g. cartesian coordinates $\leftrightarrow$ polar coordinates) without explicitly computing the map.

• A question of a different nature is whether given the solution set of some system of equations

$$\left. \begin{array}{l} f_1(x_1, \dots, x_m, y_1, \dots, y_p) = 0 \\ \vdots \\ f_p(x_1, \dots, x_m, y_1, \dots, y_p) = 0 \end{array} \right\}$$

we can (locally) solve for the y 's, i.e. if we can express each of the y_i as a function on $x_1, \dots, x_m$. This amounts to asking whether the set $S = \{f_1 = \dots = f_p = 0\} \subseteq \mathbb{R}^m \times \mathbb{R}^p$ is locally the graph of a function $G: \mathbb{R}^m \rightarrow \mathbb{R}^p$. The Implicit function theorem gives a sufficient condition for this to happen.

* Theorem (Implicit function): Let $f: P \times Q \subset V \times W \rightarrow W$ be a C^1 function, where P, Q are open. Let $S = \{(x, y) \in P \times Q : f(x, y) = 0\}$ be the solution set and let $s_0 \in S$. Let $x_1, \dots, x_m$ and $y_1, \dots, y_p$ be coordinates for V and W , respectively.

If $\det \left(\frac{\partial f_i}{\partial y_j}(s_0) \right)_{i,j=1,\dots,p} \neq 0$, then S is locally the graph of a C^1 function, i.e. there exist $X \subset P$, $Y \subset Q$ and a C^1 function $G: X \rightarrow Y$ st $S \cap (X \times Y) = \text{graph } G = \{(x, G(x)) : x \in X\}$.

Remarks: 1) Observe that the matrix $\left(\frac{\partial f_i}{\partial y_j}(s_0)\right)_{i,j=1,\dots,p}$ is the rightmost $(p \times p)$ -minor in the matrix of $f'(s_0)$, since

$$f'(s_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(s_0) & \dots & \frac{\partial f_1}{\partial x_n}(s_0) & \boxed{\frac{\partial f_1}{\partial y_1}(s_0) \dots \frac{\partial f_1}{\partial y_p}(s_0)} \\ \vdots & & \vdots & \vdots \\ \frac{\partial f_p}{\partial x_1}(s_0) & \dots & \frac{\partial f_p}{\partial x_n}(s_0) & \boxed{\frac{\partial f_p}{\partial y_1}(s_0) \dots \frac{\partial f_p}{\partial y_p}(s_0)} \end{pmatrix}$$

2) In general, when $f: \mathbb{R}^{n+p} \rightarrow \mathbb{R}^p$, one has to sort the variables in $\mathbb{R}^{n+p}$ so that the last p satisfy the requirement of the theorem.

3) By the linear version of the implicit function theorem above, the condition $\det\left(\frac{\partial f_i}{\partial y_j}(s_0)\right) \neq 0$ is equivalent to the requirement that $T_{s_0} S = \text{Ker } f'(s_0) = (f'(s_0))^{-1}(0)$ is the graph of a linear function $V \rightarrow W$.

Examples:

① Linear Algebra

Let $P_n =$ polynomials of $\deg \leq n$ with coeff. on $\mathbb{R}$, so $P_n \subseteq \mathbb{R}[x]$.

$\dim P_{n+1} = n+1$, a basis being $(1, x, x^2, \dots, x^n)$.

For $k \leq n$, $P_k \subseteq P_n$ is a linear subspace.

Consider the linear map

$$D: P_n \rightarrow P_{n-1}, \quad D(p(x)) := p'(x) \quad (\text{its derivative})$$

Matrix of D ?

$$D = \begin{matrix} & D(1) & D(x) & D(x^2) & \dots & D(x^n) \\ \begin{matrix} 1 \\ x \\ x^2 \\ \vdots \\ x^{n-1} \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & n \end{pmatrix} \end{matrix}$$

not iso because $1 \in \text{Ker } D$, so $\text{Ker } D \neq 0$.

Surjective because the minor has nonzero det, thus D has full rank.

② Linear IFT

let $L: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ given by the matrix $L = \begin{pmatrix} 2 & 3 \end{pmatrix}$

Is $L^{-1}(0) = \text{graph of linear map?}$

$$L^{-1}(0) = \text{Ker } L = \{ (x, y) \in \mathbb{R} \times \mathbb{R} : 2x + 3y = 0 \}$$

$$\Downarrow$$

$$y = -\frac{2}{3}x$$

I can solve for y

Indeed the condition of the theorem tells me the same:

$$\begin{pmatrix} 2 & \boxed{3} \\ \neq 0 \end{pmatrix}$$

Remark. I could have also solved for x,

$$2x + 3y = 0 \Leftrightarrow x = -\frac{3}{2}y$$

which agrees with the statement as $\begin{pmatrix} \boxed{2} & 3 \end{pmatrix}$

(i.e., here I took w to be the first copy of $\mathbb{R}$ in $\mathbb{R} \times \mathbb{R}$).

③ Differentiable functions

let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $f(x, y, z) = (x^2 e^z, \cos(y + z^3))$

Is it differentiable? This is composite, product and sum of C^1 functions $\Rightarrow C^1 \Rightarrow \text{diff everywhere.}$

Its derivative is

$$f'_{(x, y, z)} = \begin{pmatrix} 2x e^z & 0 & x^2 e^z \\ 0 & -\sin(y + z^3) & -\sin(y + z^3) 3z \end{pmatrix}$$

Directional derivative along $v = (1, 2, 3)$ at $p = (1, \frac{\pi}{2}, 0)$?

$$\partial_{(1, 2, 3)} f \left(1, \frac{\pi}{2}, 0 \right) = f' \left(1, \frac{\pi}{2}, 0 \right) \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

④ Inverse function theorem

Decide whether $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = (e^{x+y}, y)$

has locally an inverse around any point.

$$f'(x,y) = \begin{pmatrix} e^{x+y} & e^{x+y} \\ 0 & 1 \end{pmatrix}, \quad \det(\quad) = e^{x+y} \neq 0$$

$\Rightarrow$ Inverse FT holds! $\Rightarrow f$ has locally an inverse everywhere.

What is the derivative of the inverse at a point $f(x,y) = (e^{x+y}, y)$? By the thm,

$$(f^{-1})'(f(x,y)) = f'(x,y)^{-1}$$

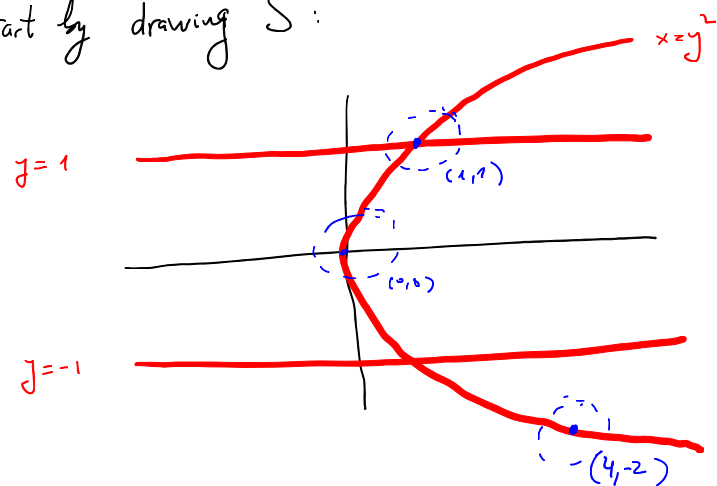
$$\begin{aligned} \begin{pmatrix} e^{x+y} & e^{x+y} \\ 0 & 1 \end{pmatrix}^{-1} &= \frac{1}{e^{x+y}} \begin{pmatrix} 1 & 0 \\ -e^{x+y} & e^{x+y} \end{pmatrix} = \\ &= \begin{pmatrix} e^{-x-y} & -1 \\ 0 & 1 \end{pmatrix} = (f^{-1})'(f(x,y)) \end{aligned}$$

⑤ Implicit function theorem

Let $S = \{(x,y) \in \mathbb{R}^2 : (y-1)(y+1)(x-y^2) = 0\} \subseteq \mathbb{R}^2$

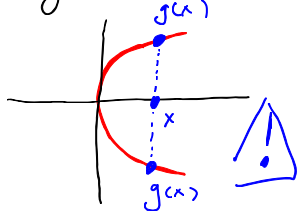
Let's study whether S is locally the graph of a function around the points $(4,-2)$, $(1,1)$, $(0,0)$. Remember that this amounts to asking whether we can solve for y (or for x) around nbhds of these points.

We start by drawing S :



Looking at the picture we already know the answer: around $(4,-2)$ we could write the circled piece ~~as~~ as the graph of a function on y ($x=y^2$) or on x ($y=-\sqrt{x}$).

Around $(0,0)$ we can solve for x ($x=y^2$) but not for y , i.e.,  is not the graph of a function $y=g(x)$ because for every x we would get two values for y !



Around $(1,1)$ we cannot solve for any of the variables as we get the same problem as above. 

Let's check our guess with the theorem:

$$f'(x,y) = \begin{pmatrix} y^2 - 1 & 2y(-2y^2 + x + 1) \end{pmatrix}$$

$$f'(4,-2) = \begin{pmatrix} 3 & 12 \end{pmatrix} \leadsto \text{we can solve for both variables}$$

$$f'(0,0) = \begin{pmatrix} -1 & 0 \end{pmatrix} \leadsto \text{can solve for } x$$

$$f'(1,1) = \begin{pmatrix} 0 & 0 \end{pmatrix} \leadsto \text{cannot solve for } x \text{ nor } y.$$